

RhinoRoFlo

ORIFICE FLOW CONTROL CHAMBER

SERF300 | SERF450 | SERF600 | DATA SHEET

SuDS
enviro
BESPOKE, STANDARDISED

OVERVIEW

RhinoRoFlo is the orifice flow control chamber in the RHINO range, sold as SERF300, SERF450 and SERF600. A removable orifice plate cap on the outlet restricts the passage of water to a pre-set maximum rate, holding back peak stormwater flow and protecting downstream drainage from overload.

Each unit is delivered factory-fitted as a one-piece HDPE twinwall chamber. A flow bypass wire and bypass plate give an integral drain-down that is operated from ground level, so the chamber can be emptied for servicing without anyone entering it.

SERF300
Ø300 mm

SERF450
Ø450 mm

SERF600
Ø600 mm

HOW THE ORIFICE WORKS

Stormwater enters through the inlet and can only leave through the orifice in the outlet cap. The orifice diameter is calculated from the site design head and the permitted discharge. As the water level rises during a storm, the chamber and upstream network hold the excess and release it at the controlled rate. A 300 mm sump sits below the outlet.

DESIGN COMPLIANCE

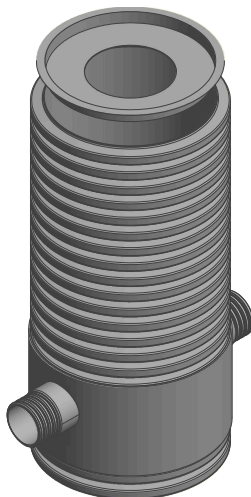
- BS EN 13598-2 (2009) plastic inspection chambers
- Sewers for Adoption 7th Edition (SfA7)
- Design and Construction Guidance (DCG), Type D and E chambers
- Building Regulations Part H1
- Adoptable and non-adoptable applications

CHAMBER OPTIONS

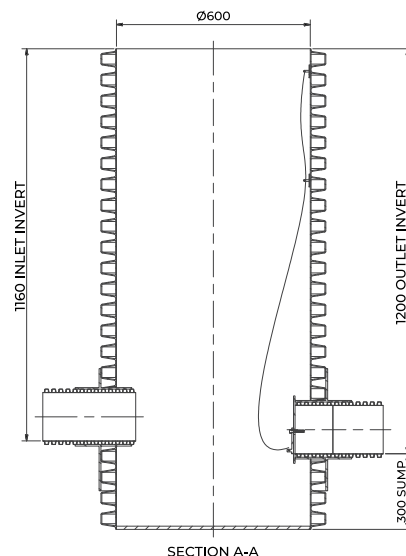
Optional reduction cap with a Ø320 mm opening for restricted access. A bespoke overflow option adds a 110 mm overflow pipe above the orifice for exceedance flows. Standard and custom sump depths are available.

INSTALLATION

Installation design should consider traffic loading, ground conditions and variable groundwater levels. If in any doubt consult the scheme designer or engineer. A suitably qualified professional is required to sign off the installation.



SERF600 with optional reduction cap.



SERF600 section A-A. Dimensions in mm from drawing SERF600150-1.5.

NOT TO SCALE

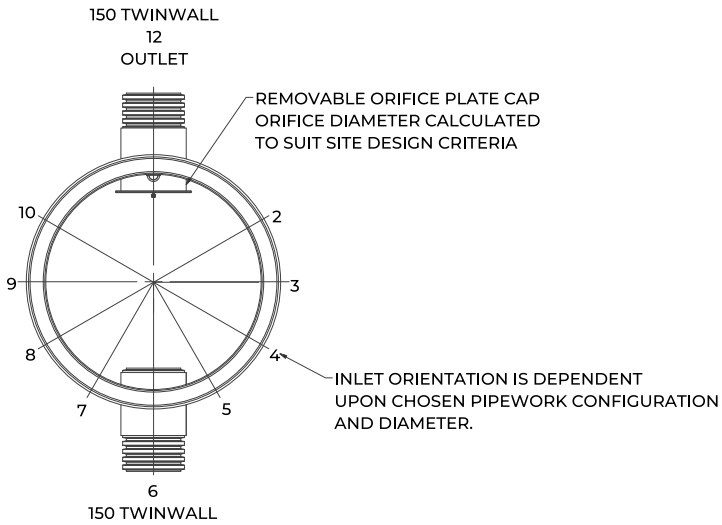
FEATURES & BENEFITS

- Orifice sized to the site design head
- Removable orifice plate cap
- Ground-level drain-down via bypass wire and plate
- Factory-fitted one-piece unit, no benching or site assembly
- Passive control, no power supply
- Flexible inlet positions around the clock face
- Thermoformed base resists fracture
- Light-duty to full-traffic loading

SPECIFICATION

Sizes	SERF300, SERF450, SERF600 (Ø300, 450, 600 mm)
Material	HDPE twinwall, extruded and thermoformed
Control	Removable orifice plate cap, orifice sized to site criteria
Drain-down	Bypass wire and plate, operated from ground level
Sump	300 mm on standard units
Depth	Adoptable up to 2000 mm, non-adoptable up to 3000 mm to pipe soffit
Design input	Design head (m)

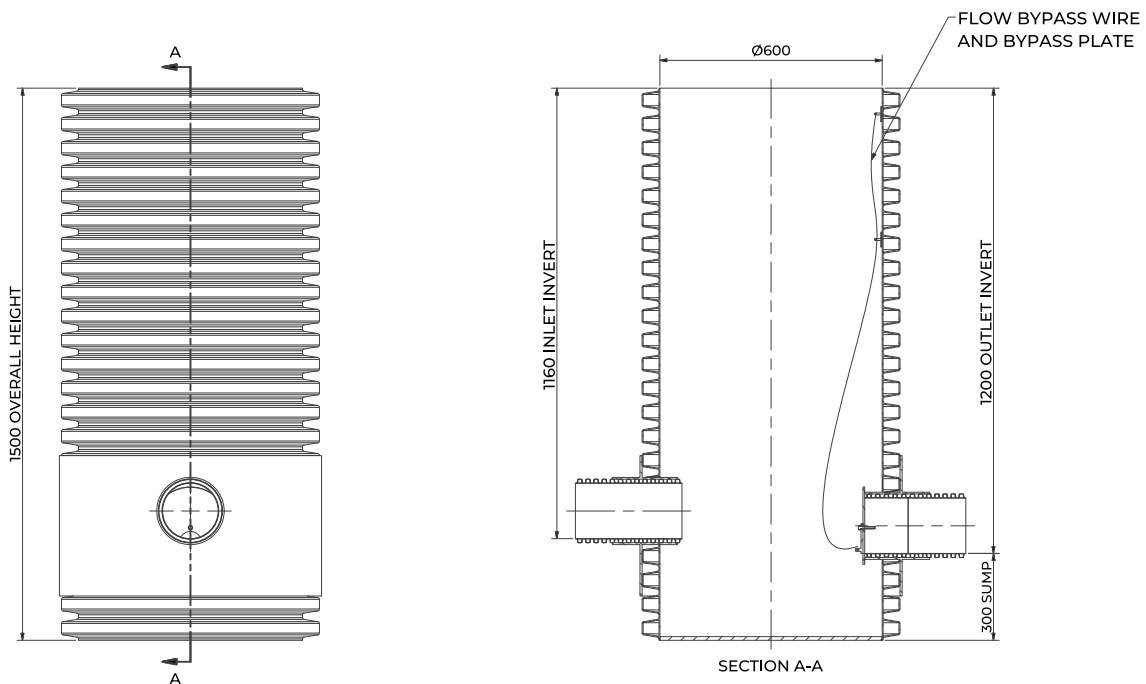
PLAN VIEW | OUTLET AT 12, INLET AT 6



SERF600150-1.5

Chamber	Ø600 mm
Overall height	1500 mm
Inlet	150 twinwall at 6 o'clock
Outlet	150 twinwall at 12 o'clock
Inlet invert	1160 mm below top
Outlet invert	1200 mm below top
Sump	300 mm
Access	Optional reduction cap, Ø320 mm opening

FRONT ELEVATION AND SECTION A-A



READING THE DRAWING

Drawing SERF600150-1.5 shows a Ø600 chamber with 150 mm twinwall inlet and outlet and a 1.5 m overall height. Invert levels are measured down from the top of the chamber.

The outlet invert sits 40 mm below the inlet invert and 300 mm above the base, leaving the sump below it.

Inlet orientation depends on the chosen pipework configuration and diameter. The orifice diameter is calculated to suit the site design criteria.

NOT TO SCALE Reproduced from the SuDS Enviro drawing pack for illustration. Dimensions in millimetres. Confirm project dimensions against the order drawing.

1 ORIFICE CONTROL

The orifice plate is fitted in a removable cap on the outlet. Its diameter is calculated for the site so that, at the design head, discharge does not exceed the permitted rate. There are no moving parts and no power supply in normal operation.

2 STORAGE AND SUMP

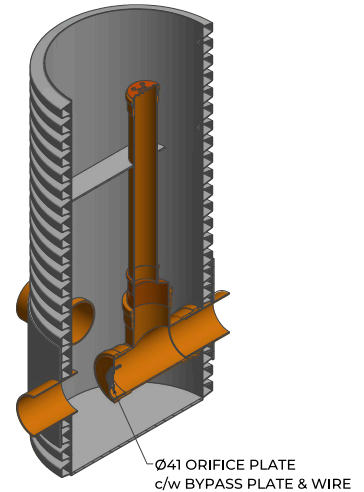
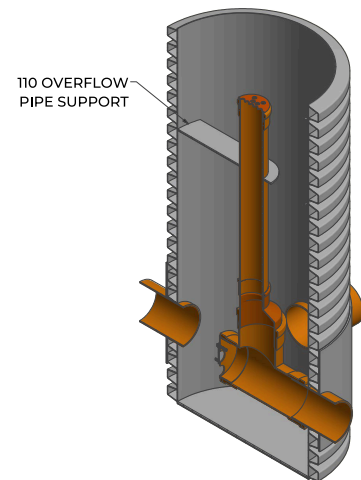
Inflow above the controlled rate backs up in the chamber and the upstream attenuation, then drains down as the storm passes. The sump below the outlet collects silt and debris so they can be removed during routine maintenance.

3 GROUND-LEVEL DRAIN-DOWN

A flow bypass wire runs from the top of the chamber down to a bypass plate at the outlet. Pulling the wire from ground level opens the bypass so the chamber can be drained for inspection or to clear a blocked orifice, without confined-space entry.

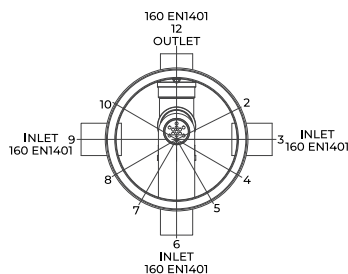
4 BESPOKE OVERFLOW OPTION

Where exceedance flows must be passed forward, a 110 mm vertical overflow pipe on a support plate rises from the outlet tee. In the example shown (SERF600160-3-6-9-1.5) the overflow top sits 135 mm below the chamber top, above a Ø41 orifice plate with bypass plate and wire.



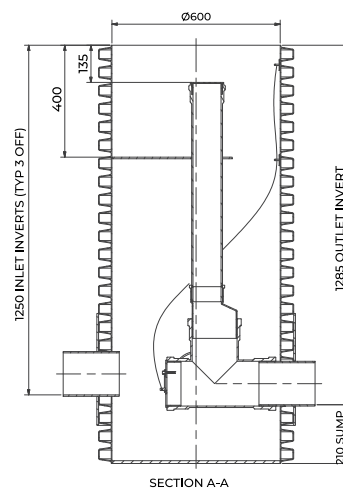
Overflow option cutaways. Drawing SERF600160-3-6-9-1.5.

PLAN | 3 INLETS AT 3, 6, 9



SERF600160-3-6-9-1.5. Ø600, 160 EN1401 outlet at 12 and three 160 EN1401 inlets. 1495 mm overall height. **NOT TO SCALE**

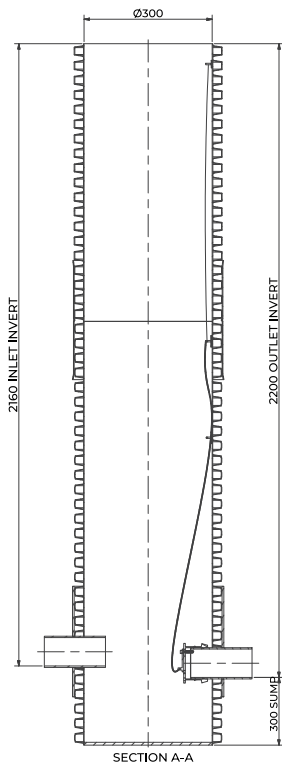
SECTION A-A



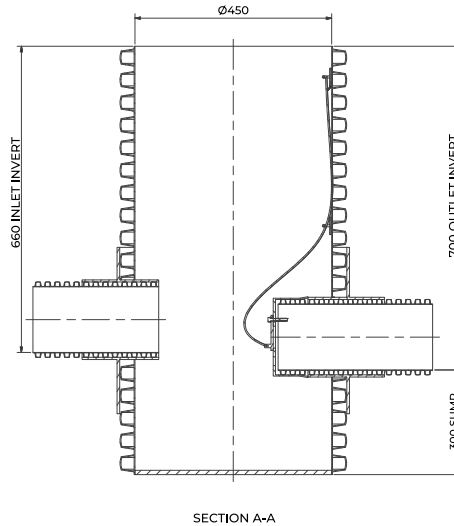
SIZING INPUTS

- Design head in metres (the only essential input)
- Permitted discharge rate, if set by consent
- Chamber size: SERF300, SERF450 or SERF600
- Inlet and outlet pipe size and type
- Inlet clock positions
- Invert depths and cover level
- Adoptable or non-adoptable
- Overflow required, yes or no

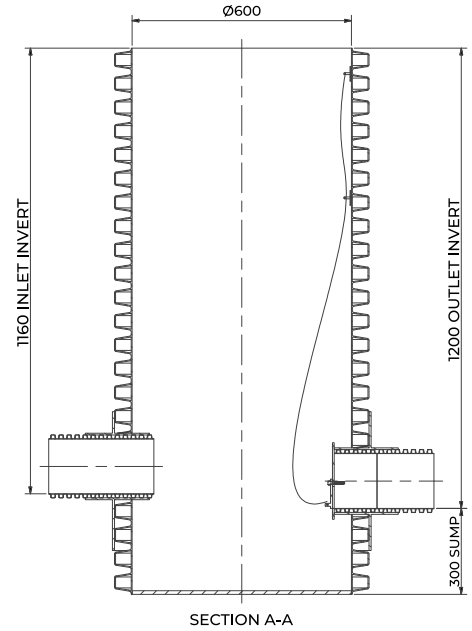
Our technical team returns a calibrated unit with the orifice plate factory-fitted and matched to the site.



SERF300 - drawing SERF300110-2.5



SERF450 - drawing SERF450150-1



SERF600 - drawing SERF600150-1.5

NOT TO SCALE Sections are shown at different scales so each fits the column. Each is one drawn configuration; overall height, pipework and inlet positions are made to order.

CODE	CHAMBER	DRAWN PIPEWORK	OVERALL HEIGHT	INLET INVERT	OUTLET INVERT	SUMP
SERF300	Ø300 mm	110 EN1401 (150 twinwall, 160 EN1401 optional)	2500 mm	2160 mm	2200 mm	300 mm
SERF450	Ø450 mm	150 twinwall	1000 mm	660 mm	700 mm	300 mm
SERF600	Ø600 mm	150 twinwall	1500 mm	1160 mm	1200 mm	300 mm

Invert levels are measured down from the top of the chamber. SERF300 is built from 300 mm twinwall with a 1 m extension and body coupler.

INSTALLATION

- Bed and surround to suit the traffic loading and ground conditions
- Allow for variable groundwater levels and flotation
- Set the outlet to the design invert; the orifice plate is factory-fitted
- Keep the bypass wire free and reachable from cover level
- Have the installation signed off by a suitably qualified professional

COMPLIANCE

- BS EN 13598-2 (2009)
- SfA7 (Sewers for Adoption)
- DCG Type D and E chambers
- Building Regulations Part H1
- DCG restricted access: max 350 mm opening over 1 m depth

SPECIFY RHINOROFLO

Send us the design head and our team will return a calibrated unit ready for site.

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